When the patrols leave, mining comes back for seconds.

What Operation Mercurio means for FZS's work in Tambopata.

Photo: giant anteater

Feb 2019	30 m	6
Operation Mercurio starts	satellite pixels since 1985	tests for cause

Anteaters eat 35,000 ants a day, and never destroy a nest

60 cm

OF TONGUE

Flicks up to 150 times a minute.

Smithsonian's National Zoo; National Geographic

1 minute

PER MOUND

Then it moves on and comes back later.

National Geographic



Photo: anteater at a termite mound

Mining in La Pampa did the same: it moved on, then came back.

THE QUESTION

What happens when an enforcement operation ends?

Operation Mercurio, La Pampa, Madre de Dios · from February 2019

WHAT WE FOUND

1 · IT FELL

1,640 → 163

ha/yr net new mining, Tambopata buffer zone

2 · IT MOVED

23% → 65%

of new mining near La Pampa north of the road

3 · IT CAME BACK

1,829

ha/yr in 2022–25, much of it on old pits

All three: description, 2016–18 vs 2019–21 vs 2022–25.

DESCRIPTION

La Pampa: one road away from land where mining is legal

1 **Protected areas**

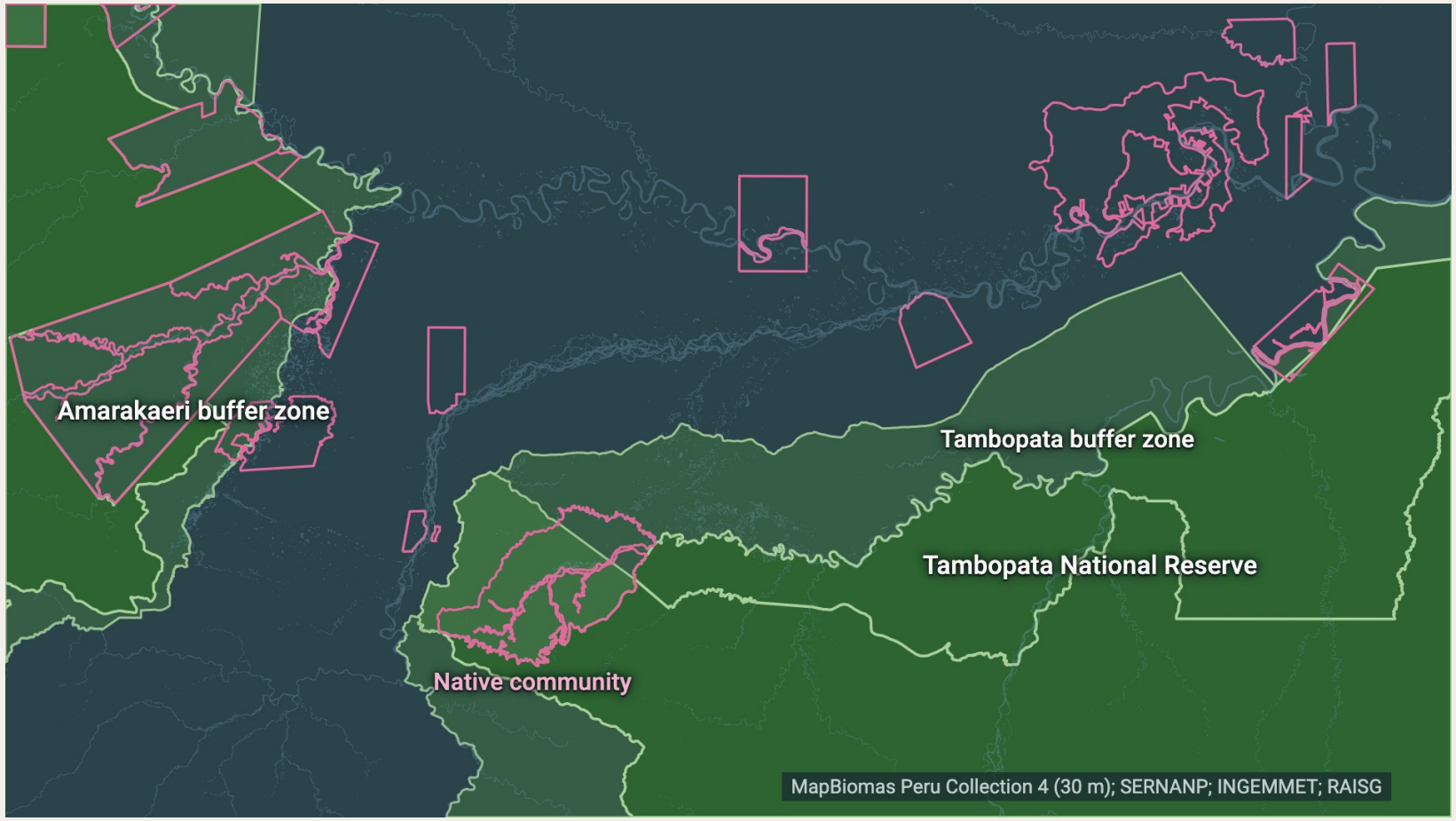
Tambopata reserve, its buffer zone, Amarakaeri. Pink: native land.

2 The Interoceanic Highway

3 Rivers and old mining

4 Where mining can be legal

5 The target



DESCRIPTION

La Pampa: one road away from land where mining is legal

1 Protected areas

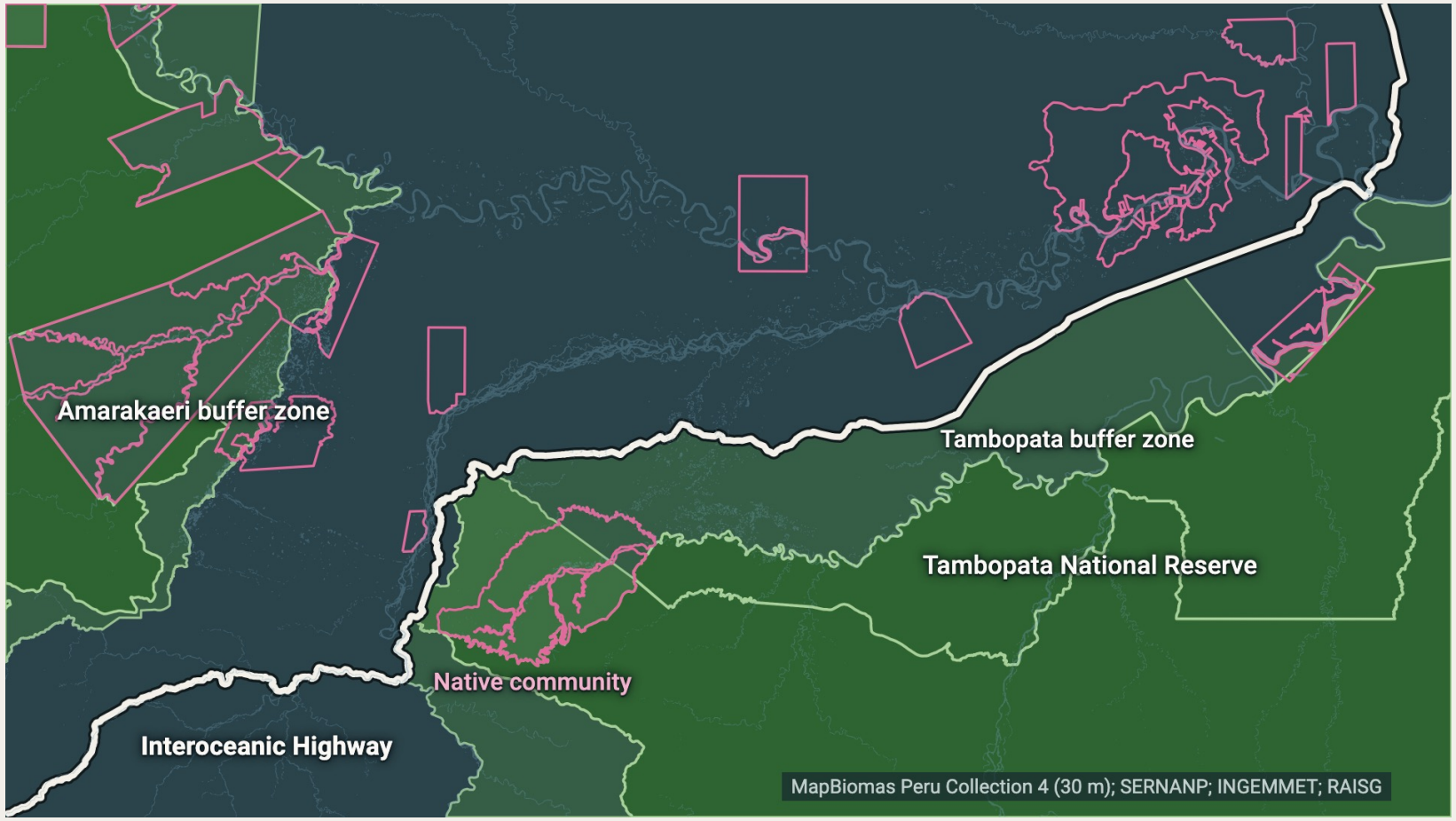
2 **The Interoceanic Highway**

One road splits the buffer zone from the land to the north.

3 Rivers and old mining

4 Where mining can be legal

5 The target

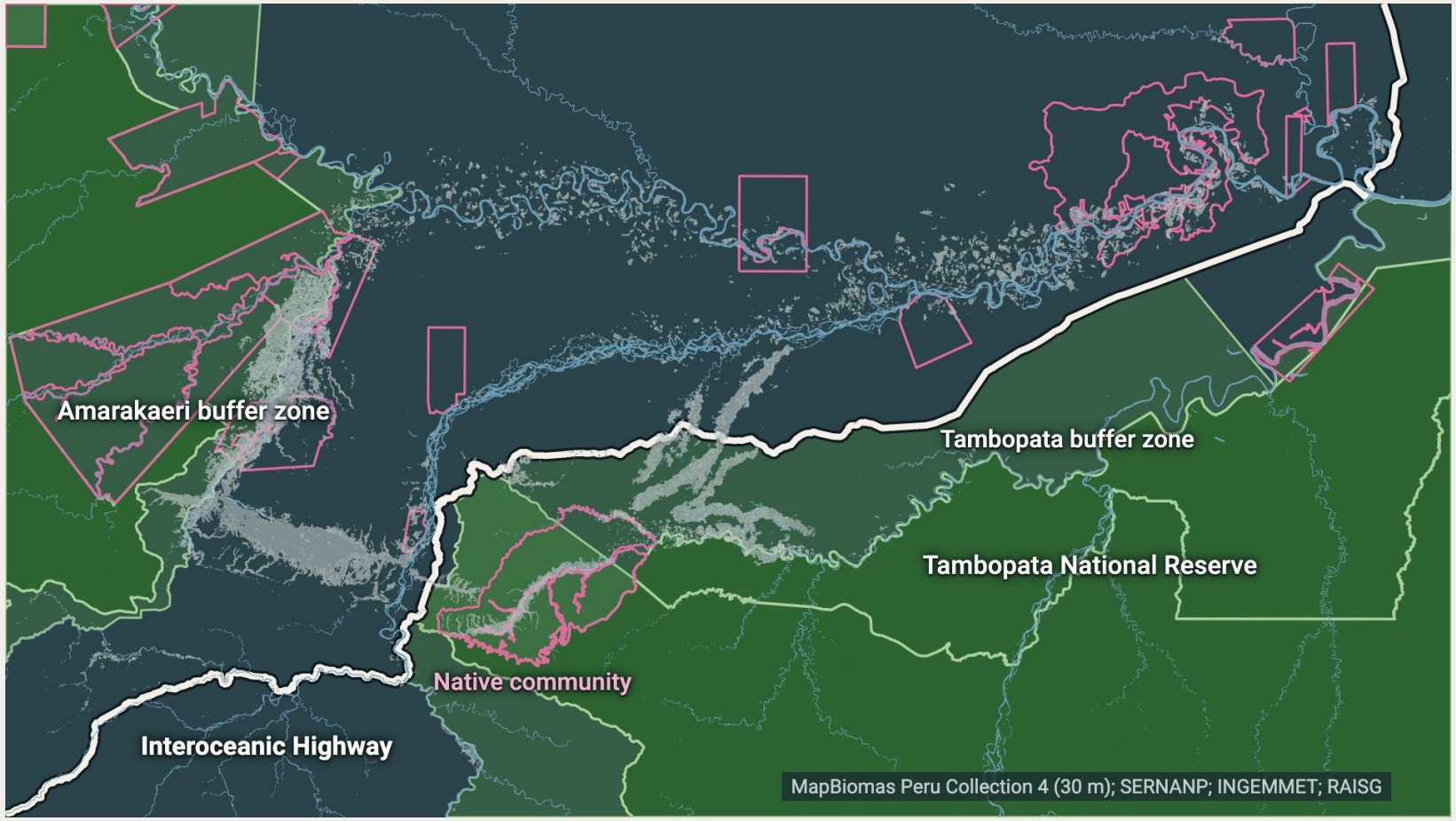


DESCRIPTION

La Pampa: one road away from land where mining is legal

1. Protected areas
2. The Interoceanic Highway
3. **Rivers and old mining**

   Alluvial gold sits in floodplains. Grey: mined by 2018.
4. Where mining can be legal
5. The target

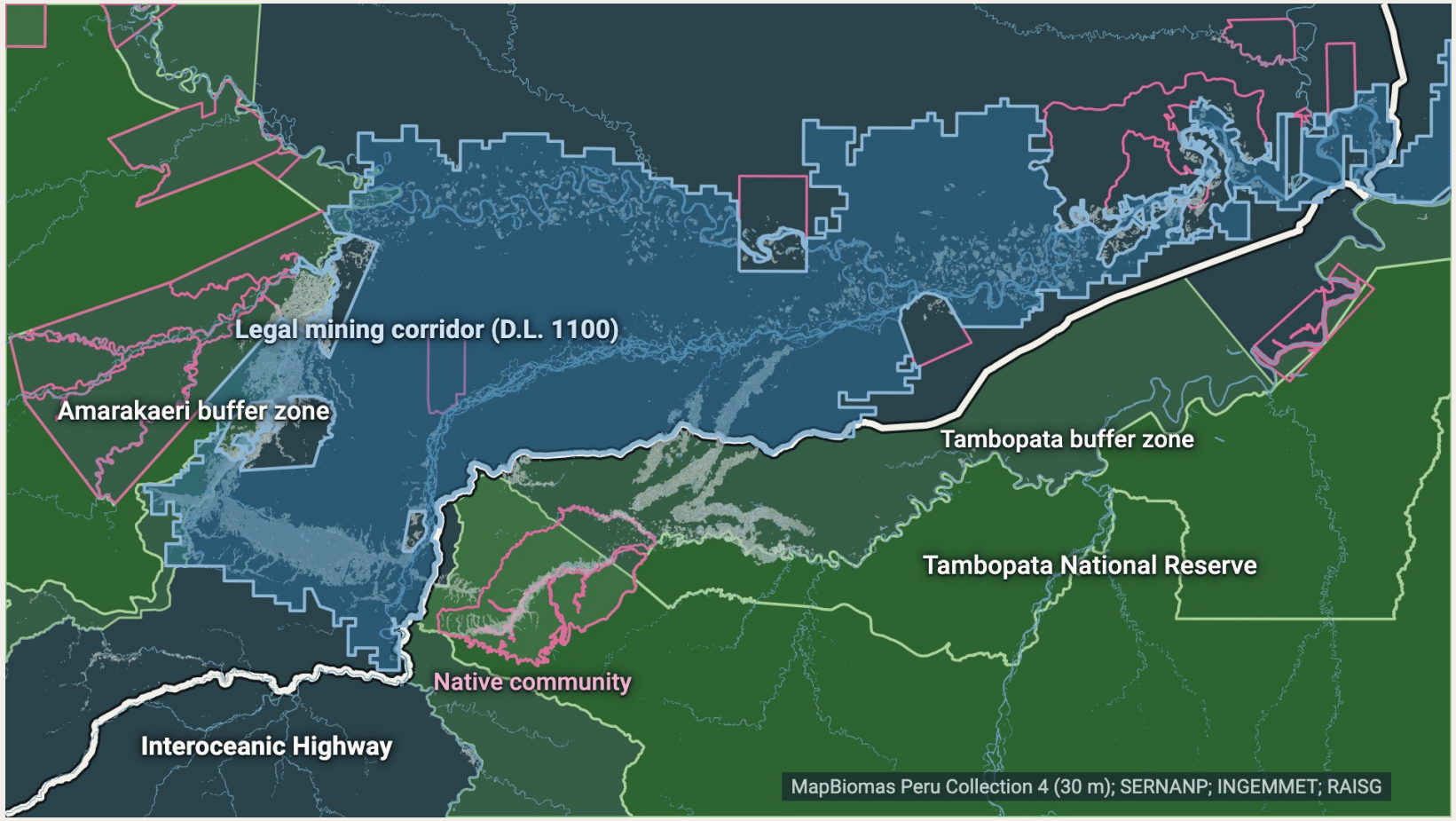


DESCRIPTION

La Pampa: one road away from land where mining is legal

1. Protected areas
2. The Interoceanic Highway
3. Rivers and old mining
4. **Where mining can be legal**

   D.L. 1100 corridor, north of the road.
   South of it: a crime.
5. The target

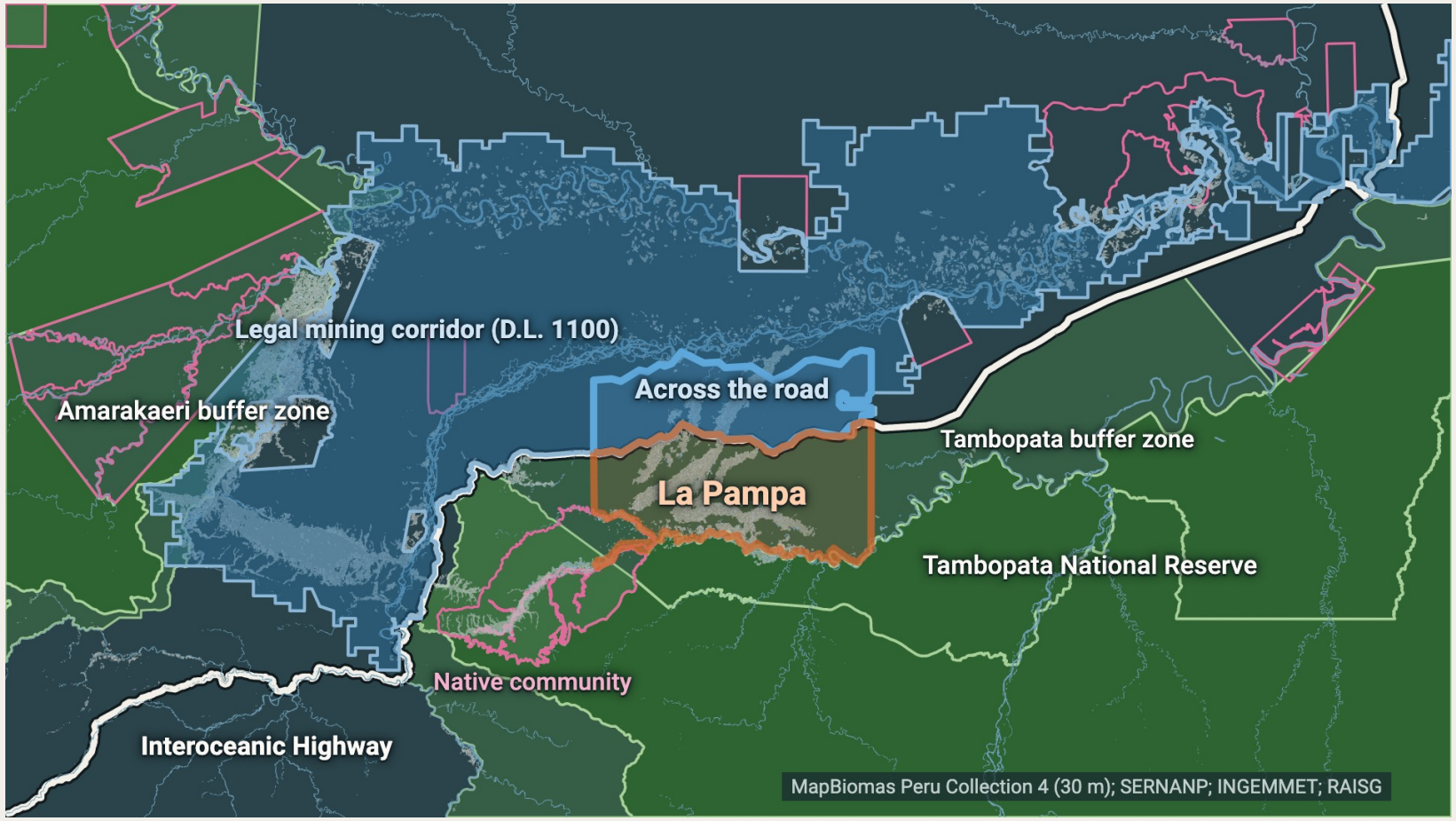


DESCRIPTION

La Pampa: one road away from land where mining is legal

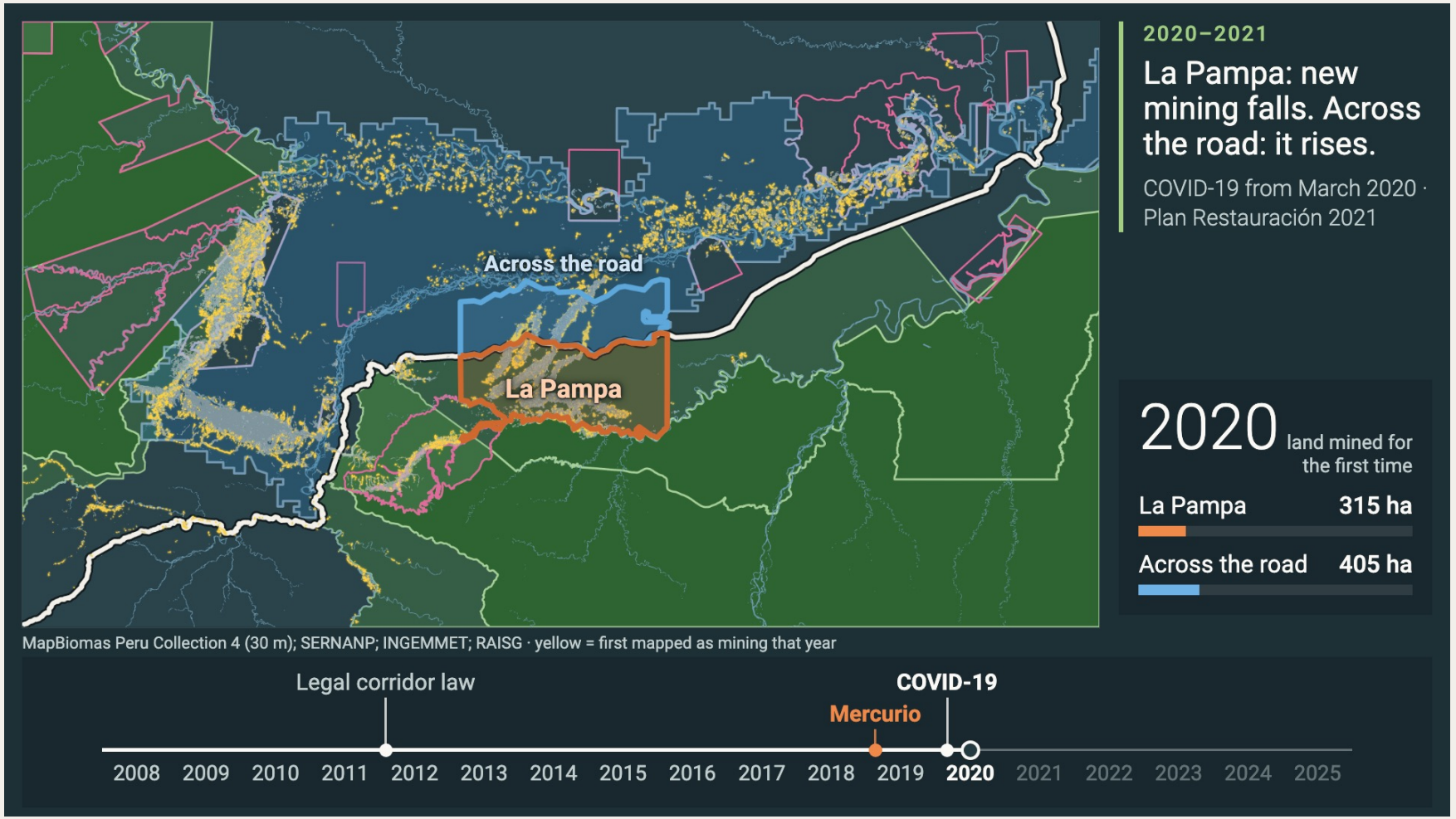
1. Protected areas
2. The Interoceanic Highway
3. Rivers and old mining
4. Where mining can be legal
5. **The target**

   La Pampa, target of Operation Mercurio from Feb 2019.



DESCRIPTION

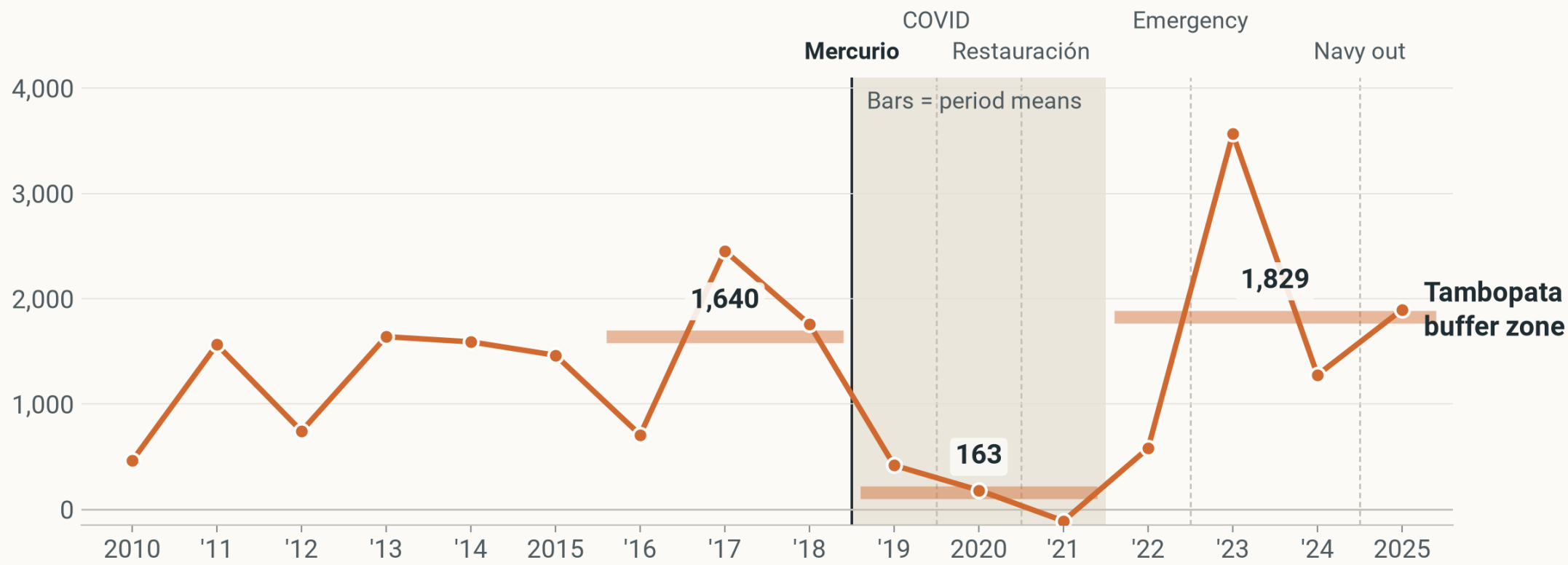
Where land was mined for the first time, 2008–2025



DESCRIPTION

Net new mining in Tambopata fell 90%, then came back

Tambopata buffer zone · net new mining, ha per year · bars = period averages

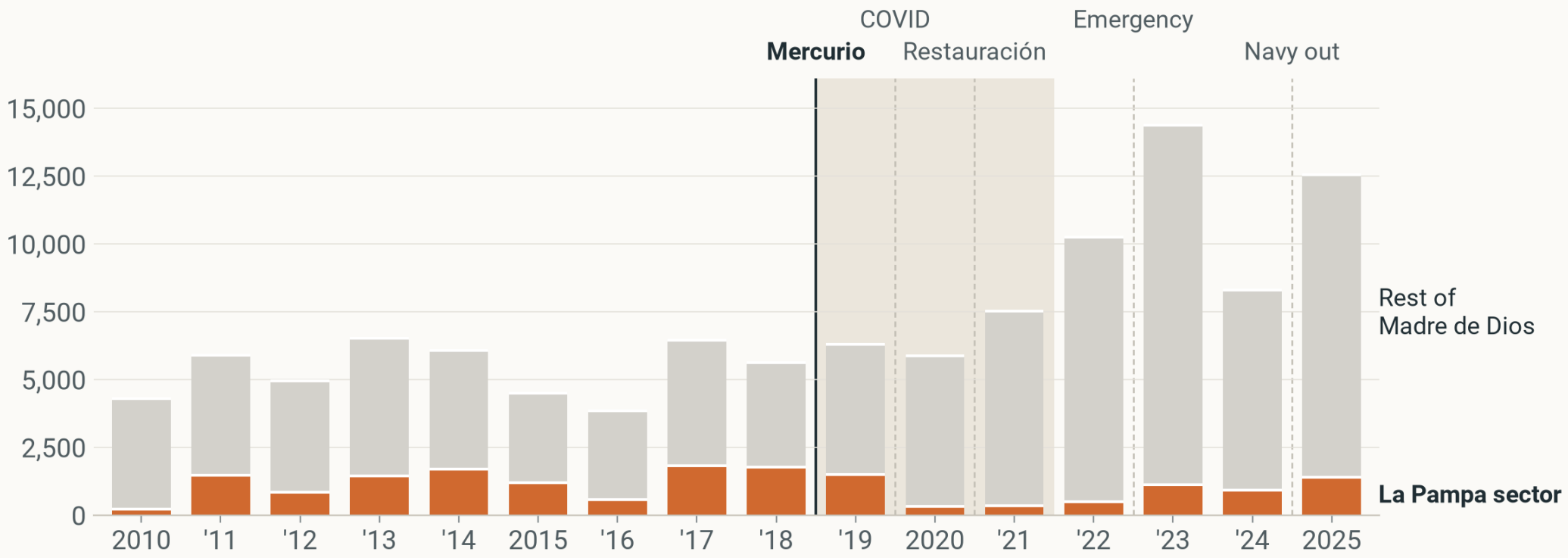


Source: MapBiomas Peru, Collection 4 (class 4.2 Minería), course table peru_bufferzone_year.csv (all department rows summed). Net additions can hide returns to old pits. Satellites cannot see river dredges or mercury.

DESCRIPTION

La Pampa: first-time mining halved in 2019–21

Land mined for the first time, ha per year · orange = La Pampa, grey = rest of Madre de Dios

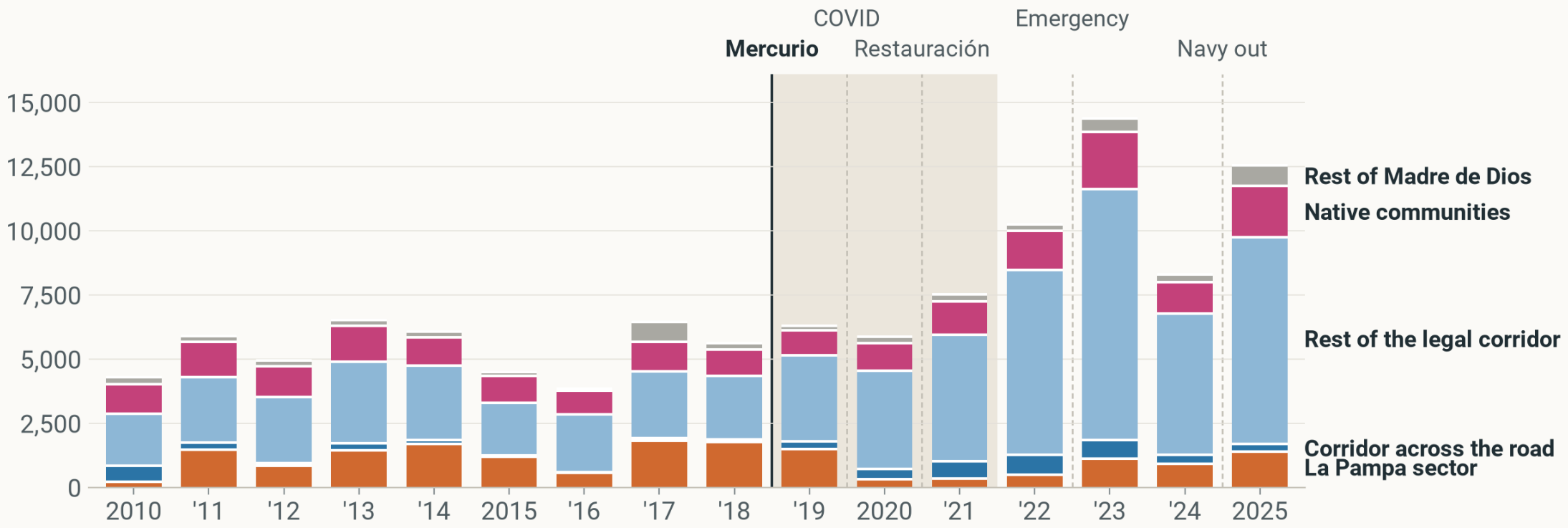


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. Legality is by location; the map class can't tell legal from illegal mining. Satellites cannot see river dredges or mercury.

DESCRIPTION

The region kept growing, mostly in the legal corridor

All Madre de Dios, first-time mining: 5,317 → 6,571 → 11,373 ha/yr

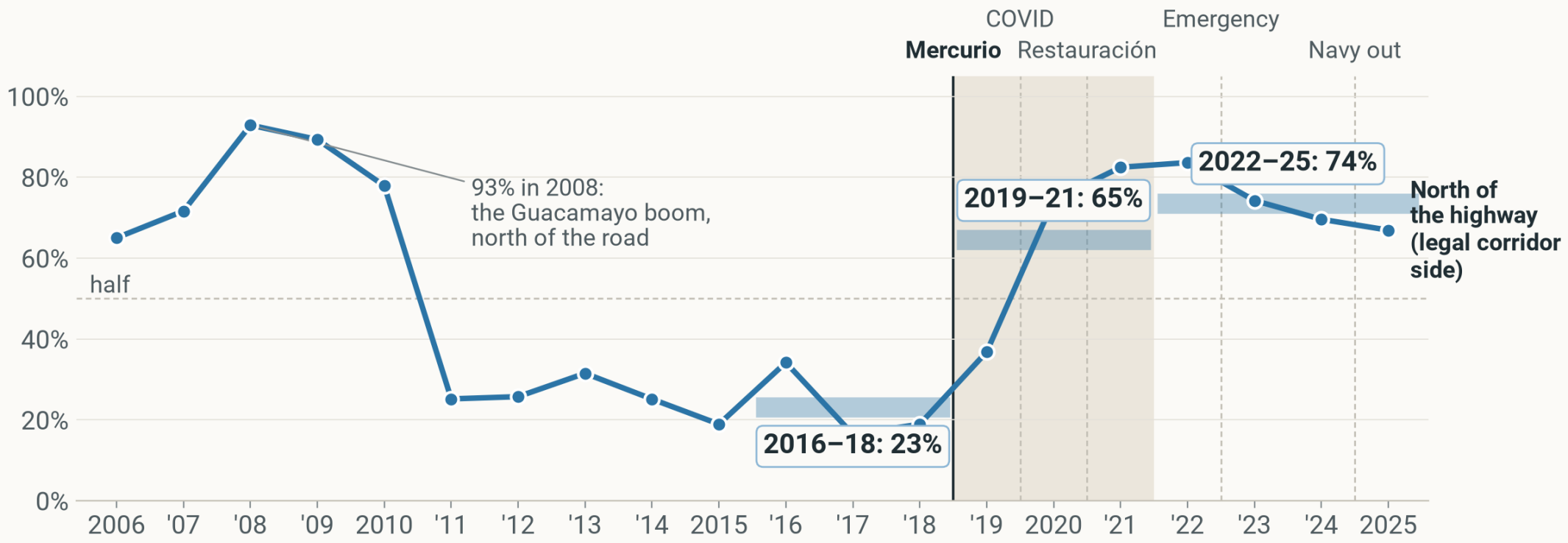


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. Legality is by location; the map class can't tell legal from illegal mining. Satellites cannot see river dredges or mercury.

DESCRIPTION

New mining near La Pampa moved north of the road: 23% → 65%

Share of first-time mining north of the Interoceanic Highway, La Pampa's longitudes

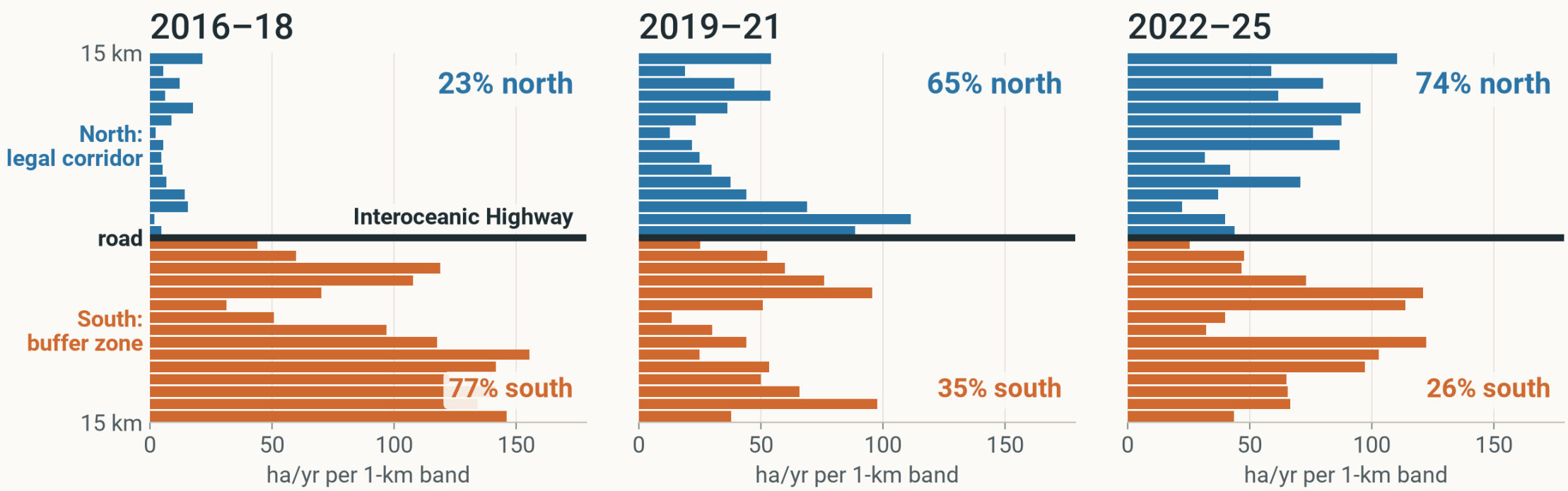


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. North = legal corridor side; south = Tambopata buffer zone. Satellites cannot see river dredges or mercury.

DESCRIPTION

In 2019–21 new mining piled up just north of the road

First-time mining per 1-km strip, ha/yr · blue = north (legal side), orange = south (buffer zone)

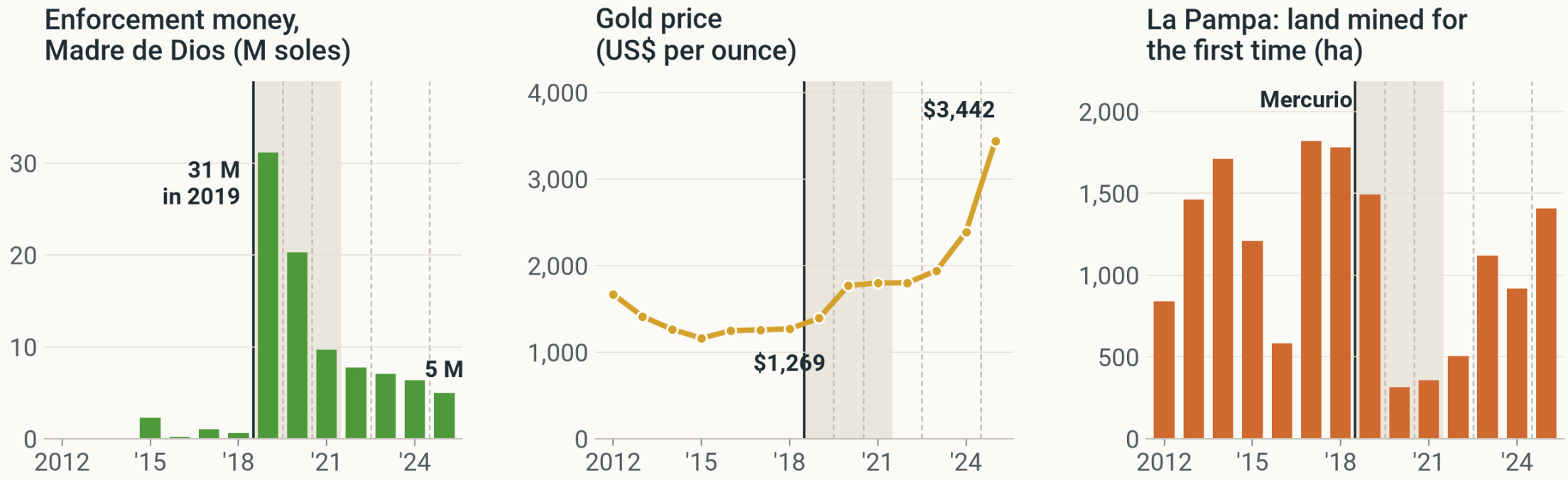


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. Bands shown to ±15 km; shares use all bands to ±30 km. Satellites cannot see river dredges or mercury.

DESCRIPTION

Money faded after 2019, while gold rose 2.7-fold

Enforcement spending, Madre de Dios · gold price · La Pampa first-time mining

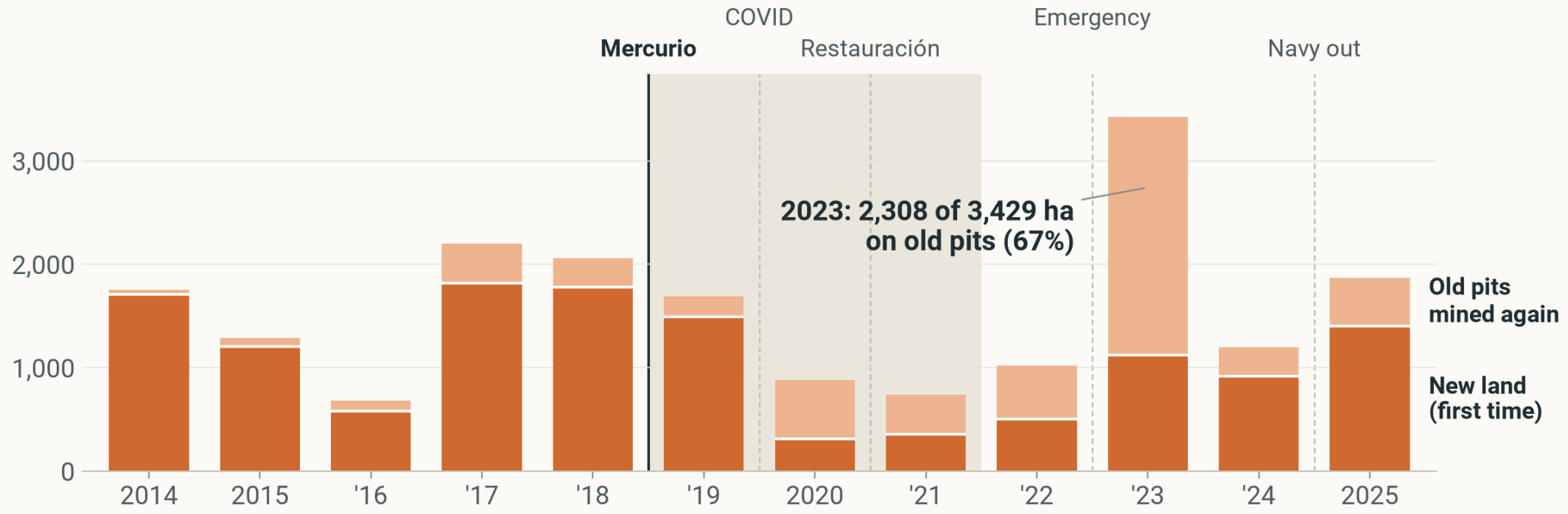


Source: MEF Consulta Amigable, PP 0128 executed spending (devengado), Madre de Dios; World Bank Pink Sheet (nominal); MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. Correlation is not shown: 11 years, and money follows mining.

DESCRIPTION

In 2023, 67% of new mining in La Pampa was on old pits

La Pampa, ha per year · dark = new land · light = old pits mined again

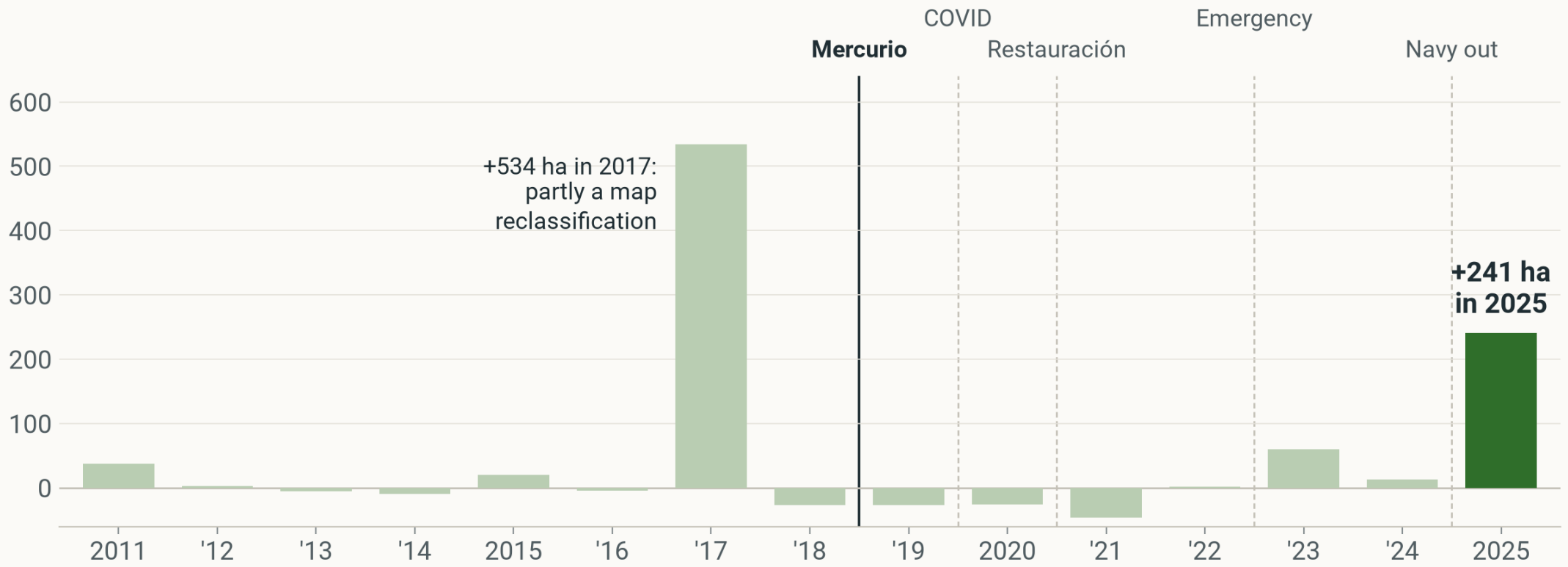


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), cut to our zones (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; legal corridor = D.L. 1100, INGEMMET). First-time mining = pixels mapped as mining for the first time since 1985. Old-pit counts include some year-to-year map flicker (252 ha/yr already in 2016–18): trust the 2023 jump, not small changes. MAAP #193 (2023) reports miners back on degraded land. Satellites cannot see river dredges or mercury.

DESCRIPTION

In 2025, mining inside the reserve jumped by 241 ha

Tambopata National Reserve · net new mining, ha per year



Source: MapBiomas Peru Collection 4, protected-area statistics (class 4.2 Minería; platform check 777 ha, 2025). MAAP #241 (Pacsi et al., 2026) reports ~500 ha inside the reserve, late 2025 to Feb 2026. Satellites cannot see river dredges or mercury.

CAUSAL · CONDITIONAL

We tried six methods. Two held up

Each builds a stand-in for Tambopata without Mercurio



Difference-in-differences

vs 14 other Peruvian buffer zones

HELD UP · IN HECTARES



Synthetic control

a weighted copy of Tambopata

NO GOOD COPY



Rule-based twins

look-alike areas in Brazil

DIFFERENT TRENDS BEFORE 2019



Spatial discontinuity

just north vs just south of the road

BORDERLINE · P = 0.08



Matching, 7 ways

areas picked by algorithm

CAN'T BEAT CHANCE



Synthetic DiD

weighted zones and years

HELD UP · IN HECTARES

CAUSAL · CONDITIONAL

What we can say about cause

If the other zones show what Tambopata would have done, Mercurio and the operations after it cut net new mining by 411 to 1,859 ha a year in 2019–21.

Hectares only

Largest fall of 15 zones; for its size, 3rd of 7.

p = 0.07

The best possible with one target among 15.

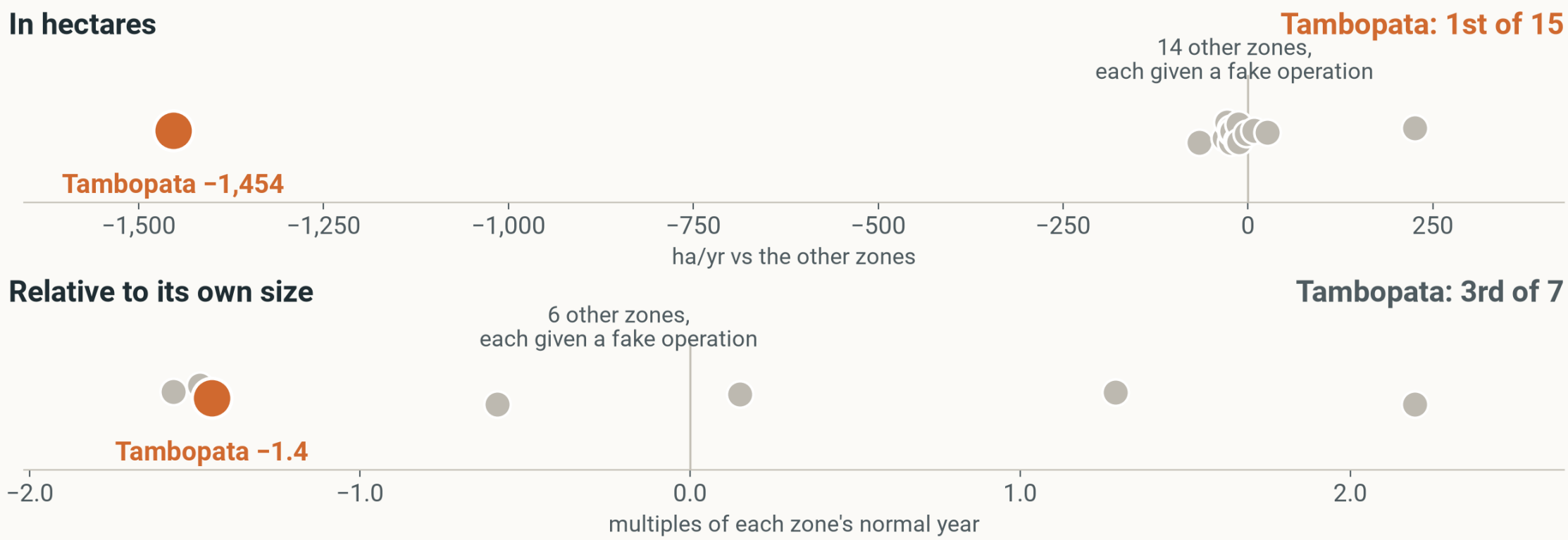
COVID overlaps

Nothing fell in 2019. The fall came in 2020–21.

CAUSAL · CONDITIONAL

In hectares the fall stands out. For its size, it doesn't

Orange = Tambopata's real 2019–21 change · grey = other zones given a fake operation



Source: MapBiomas Peru Collection 4, peru_bufferzone_year.csv; 15 Peruvian buffer zones with mining before 2019 (scaled: 7 with > 10 ha/yr). Tambopata removed from each fake zone's comparison pool. Placebo p = rank / n.

DESCRIPTION

Most biases make the fall look bigger

SATELLITES CAN'T SEE

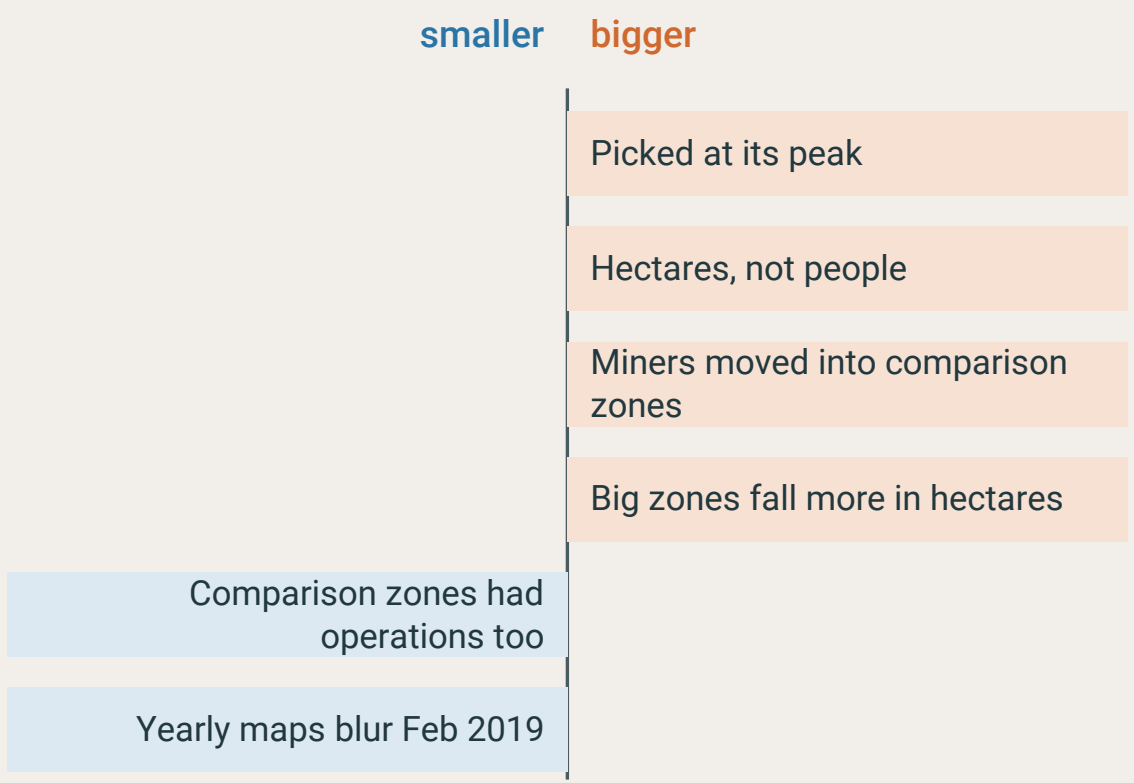
River dredges
148 → 598 in La Pampa, 2021–23 (MAAP #193)

Mercury
in rivers and fish

People
hectares, not miners

Legality
legal and illegal look alike

EACH BIAS MAKES THE 2019–21 FALL LOOK…



Read the fall as an upper bound, the return as a lower bound.

RECOMMENDATION

Plan for the months after the operation ends

Stay after the sweep

2019–21

the only years with a fall. Fund lasting presence, not one-off raids.

Guard the reserve's edge and river

91%

of new mining inside the reserve is within 1 km of its edge.

Watch the legal side and old pits

69 → 463

ha/yr of new mining just across the road.

Questions?

Like the anteater, we'll come back for more.



Photo: rangers / Tambopata river

For questions

Methods · estimates · placebos · timing · depth · legality · declared gold · money · risk map · Brazil · map video

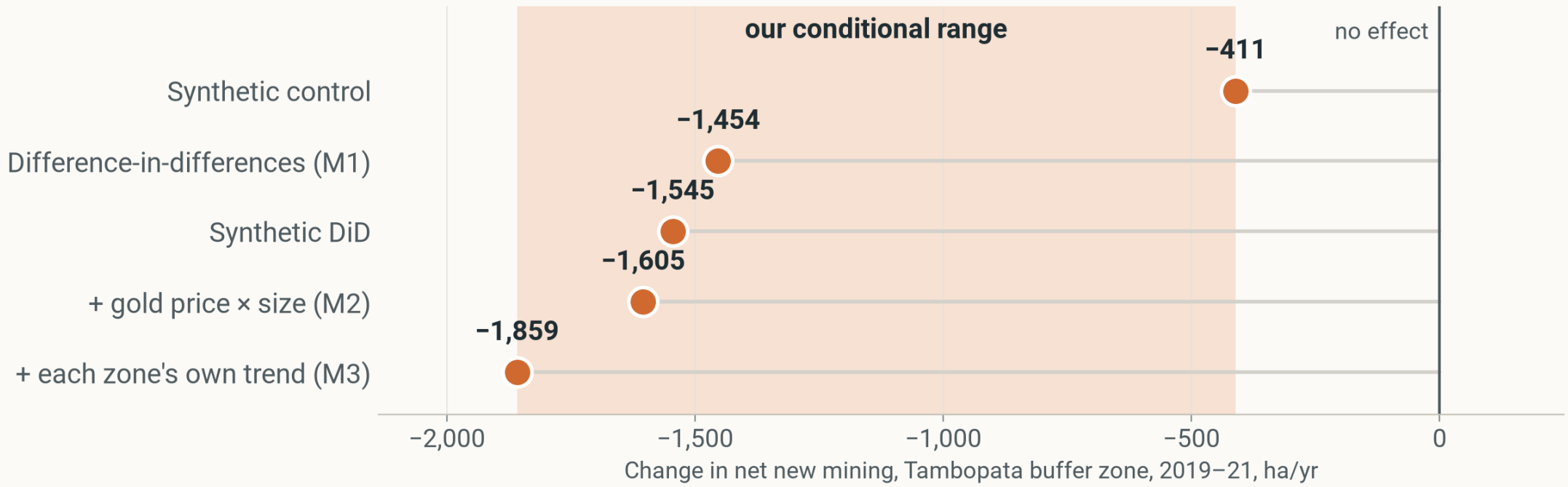
Six methods in detail

METHOD	COMPARES TAMBOPATA WITH	RESULT	VERDICT
1 Difference-in-differences	14 Peruvian buffer zones	-1,454 to -1,859 ha/yr; rank 1 of 15	Held up
2 Synthetic control	Weighted mix of zones	-411 ha/yr; no pre-2019 fit	Weak
3 Rule-based twins	Amanã and Tapajós (Brazil)	40% of the twins' pace; different pre-trends	Weak
4 Spatial discontinuity	Either side of the road	+2.1 pts north (p = 0.08)	Borderline
5 Matching (7 methods)	Areas picked by algorithm	All slower; placebo p 0.19–0.57	Can't decide
6 Synthetic DiD	Weighted, trend-matched zones	-1,545 ha/yr; scaled rank 3 of 7	Held up

CAUSAL · CONDITIONAL

Five estimates, one conditional range

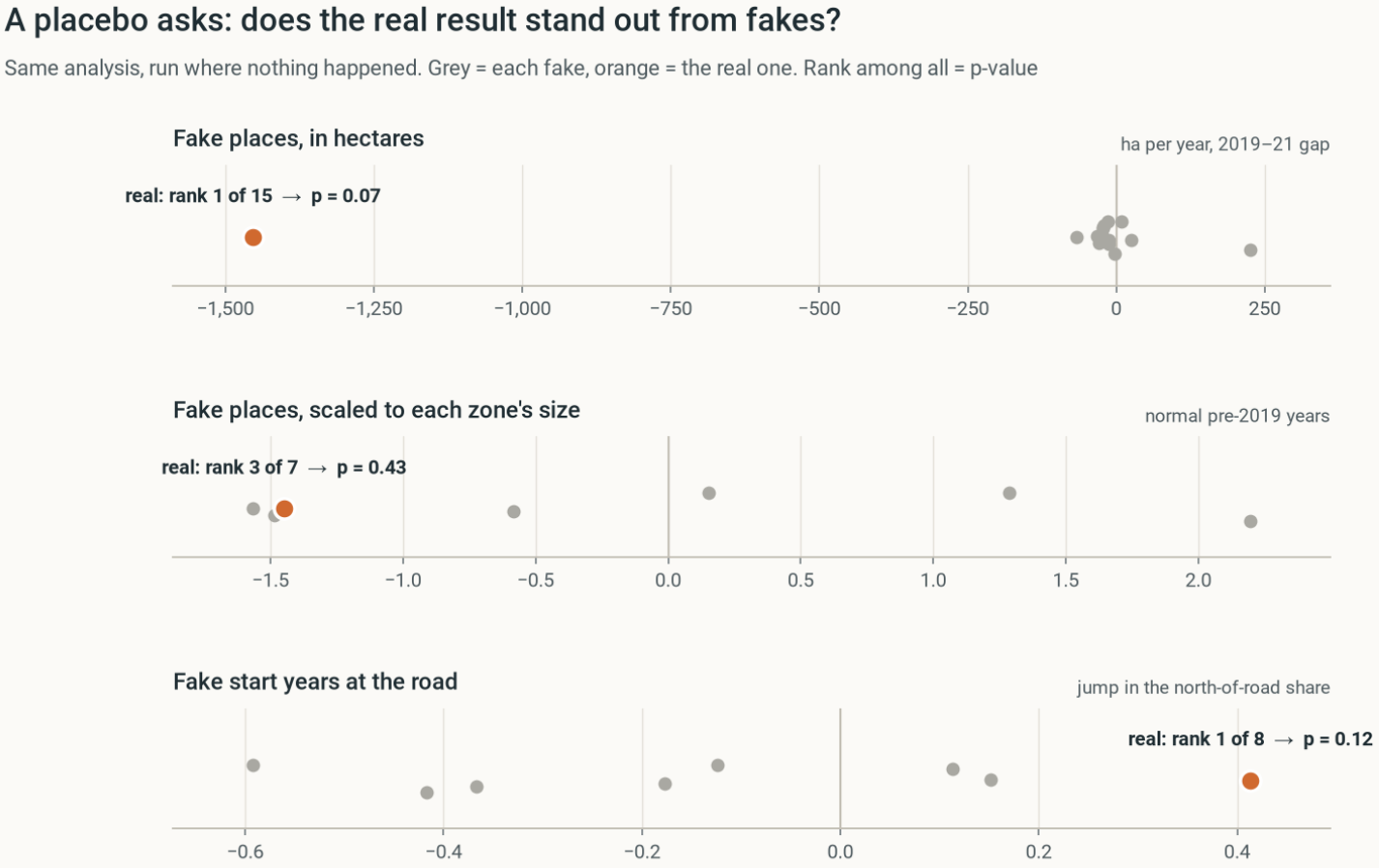
Change in net new mining, Tambopata buffer zone, 2019–21, ha/yr



Source: MapBiomas Peru Collection 4, peru_bufferzone_year.csv; zone + year effects (M1–M3), synthetic control and synthetic DiD on the same 15-zone panel (peru_models.py, peru_sdid.py).

DESCRIPTION

How a fake test shows if the real result stands out

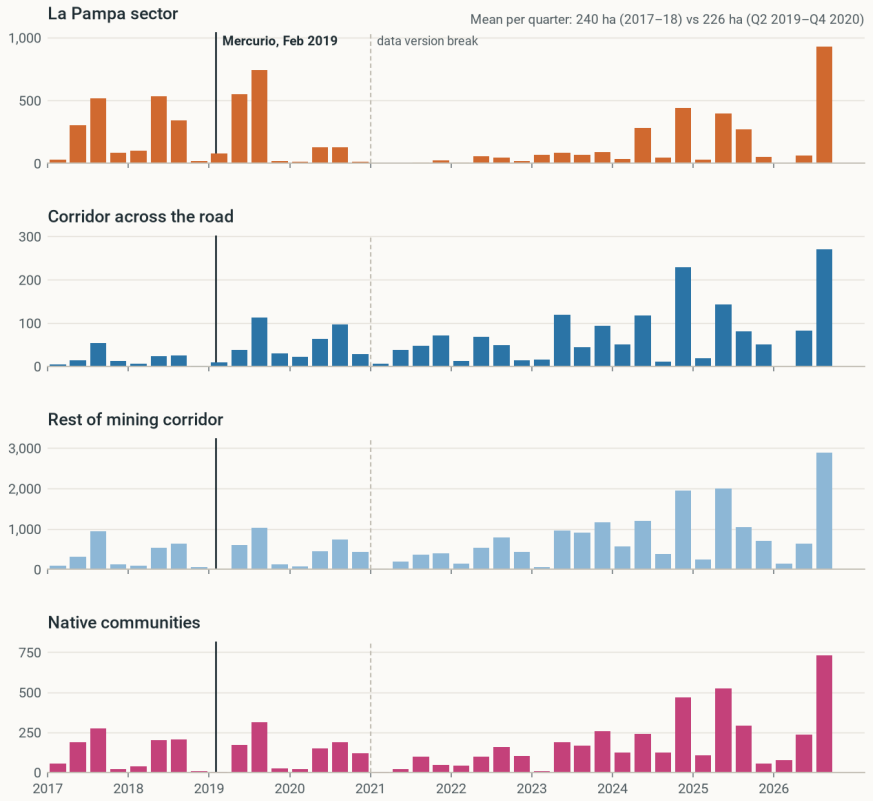


DESCRIPTION

Clearing in La Pampa went on through 2019

La Pampa kept clearing forest through 2019. Across the road, clearing rose from mid-2019.

Forest-loss alerts (GLAD-L) on mapped mining land, hectares per quarter. Each panel has its own scale.



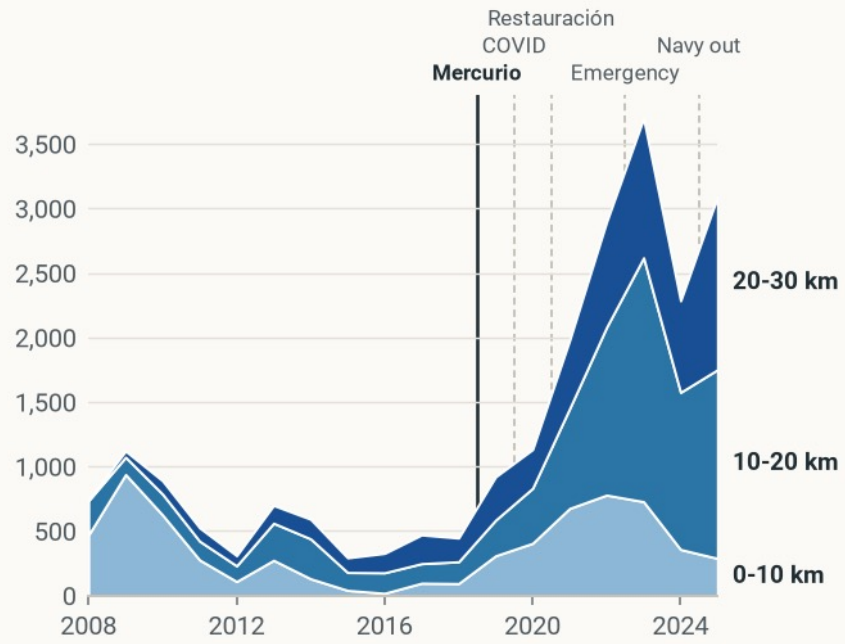
Source: UMD GLAD-L forest-loss alerts (Landsat; C1 2017-20, C2 2021-26) x MapBiomas Peru Col 4 mining mask x zones v2. Alerts bunch in the dry season (Q3) because of clouds; compare the same quarter across years. Alerts need standing forest, so returns to old pits never appear. Description.

New mining hugged the road, then spread deeper

After 2019 new mining north of La Pampa started nearer the road, then pushed deeper

First-time mining north of the highway, La Pampa longitudes, 2008–2025

Hectares per year, by distance band



Mean distance north of the highway, km



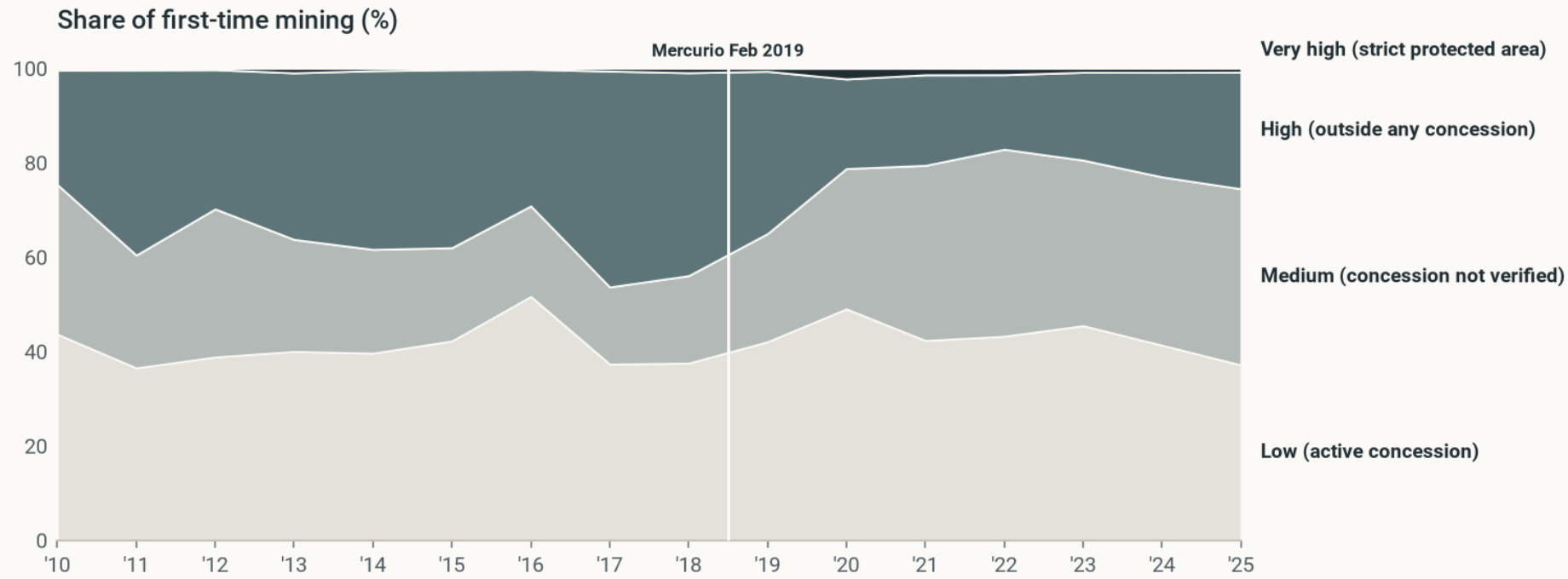
Source: MapBiomas Peru Col 4 rasters, zones v2. La Pampa longitudes only; first-time mining (new land). Break test: OLS of mean distance on a trend and a post-2019 dummy, Newey-West errors, 2008–2025 (18 years). Description.
Vertical lines: an event in year Y is drawn at Y - 0.5, just before the first annual addition that can reflect it (addition(Y) = map(Y) - map(Y-1)).

DESCRIPTION

Presumed-illegal new mining fell; grey zones grew

Presumed-illegal new mining fell from 41% to 21% of the total

First-time mining in Madre de Dios by MAAP legality level, share of each year's hectares (%). High + Very high: 41% → 25% → 21%



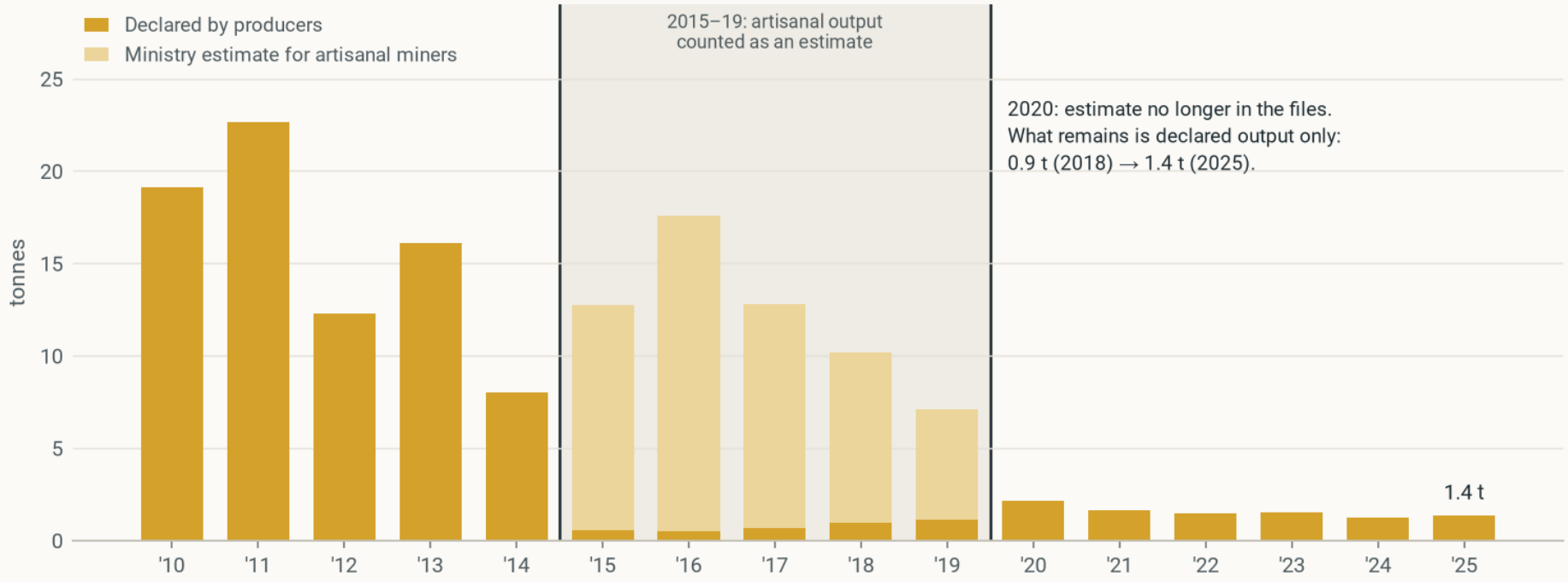
Source: Amazon Mining Watch / MAAP #235 legality layer (GitHub earthrise-media/mining-detector) x MapBiomas Peru Col 4 first-mined year. Today's concessions applied to all years (a time-consistent version using filing dates gives the same fall). Description.

DESCRIPTION

The official gold drop is a definition change

Official gold 'fell' in 2020 because the ministry stopped adding its artisanal estimate

Madre de Dios gold in the official series, tonnes per year: what producers declared plus MINEM's estimate



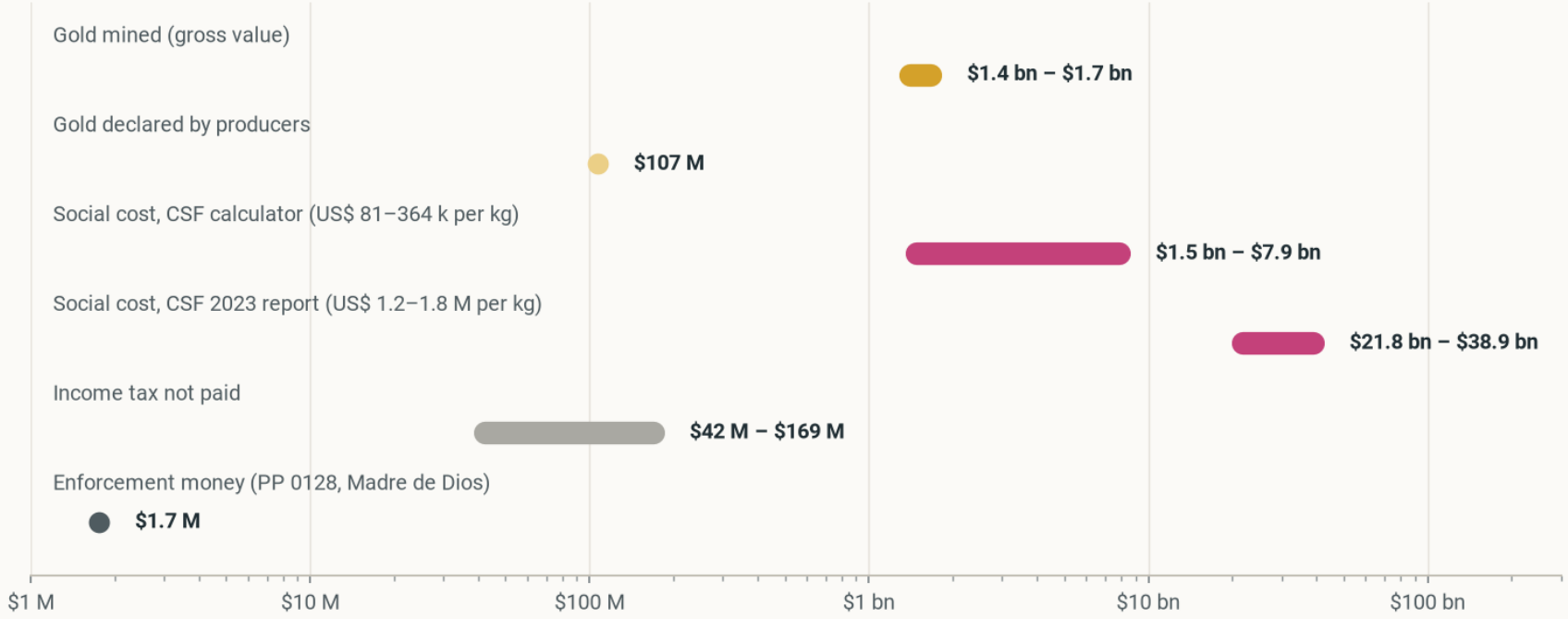
Source: MINEM, Producción minera (gob.pe 5472883), yearly gold files; Madre de Dios rows. Total of both parts = the central bank series (checked for 2012 and 2019). Description.

PREDICTION

Gold and its damage dwarf enforcement money

Each year, mining's damage and gold dwarf the money spent stopping it

Madre de Dios, per year, 2022–25, US$ (log scale). Bars = low–high range from cited parameters. Informed speculation, not measurement

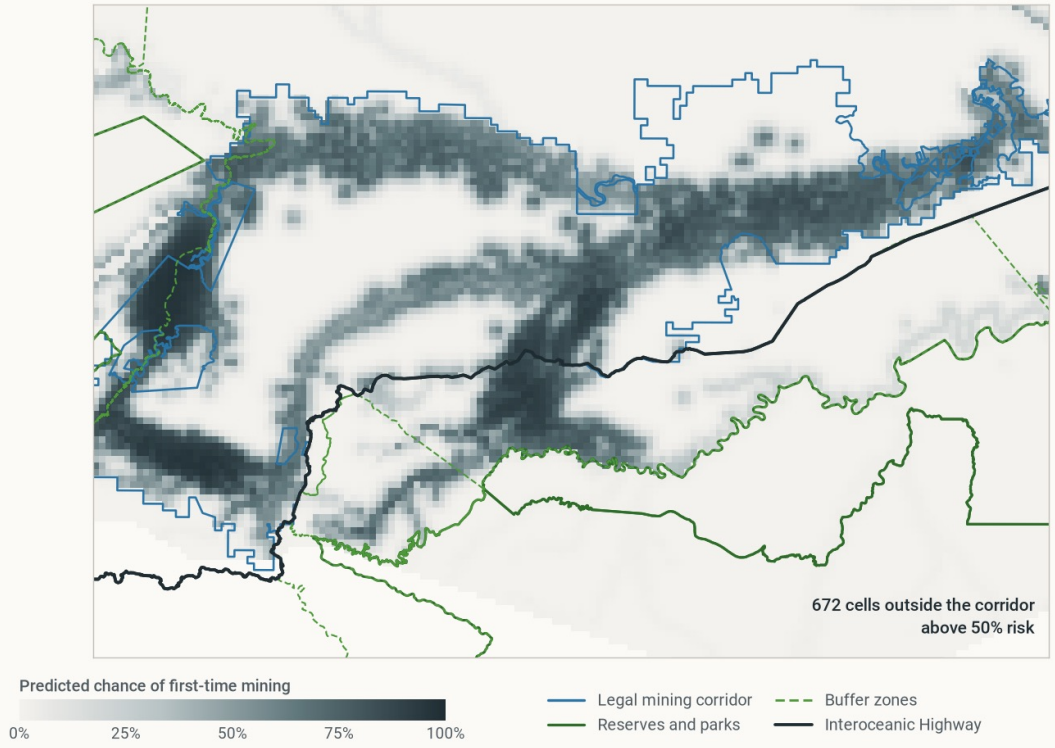


Sources: our first-time mined hectares (MapBiomas Peru Col 4; old pits given no gold) x CSF (2023) 1.6–1.9 kg gold/ha; World Bank gold price; MINEM declared gold; income tax 29.5% x 11–37% margin (D.L. 1269; Torres 2015); social cost: two CSF estimates shown separately; MEF PP 0128 at BCRP annual exchange rates. Full citations: docs/overnight/notes/economics_sources.md.

Next new mining is likely next to old mining

New mining is predicted where mining already is

Chance of first-time mining in 2026–28 per 1-km cell. Held-out AUC 0.92, vs 0.91 for 'mining nearby' alone. Prediction.

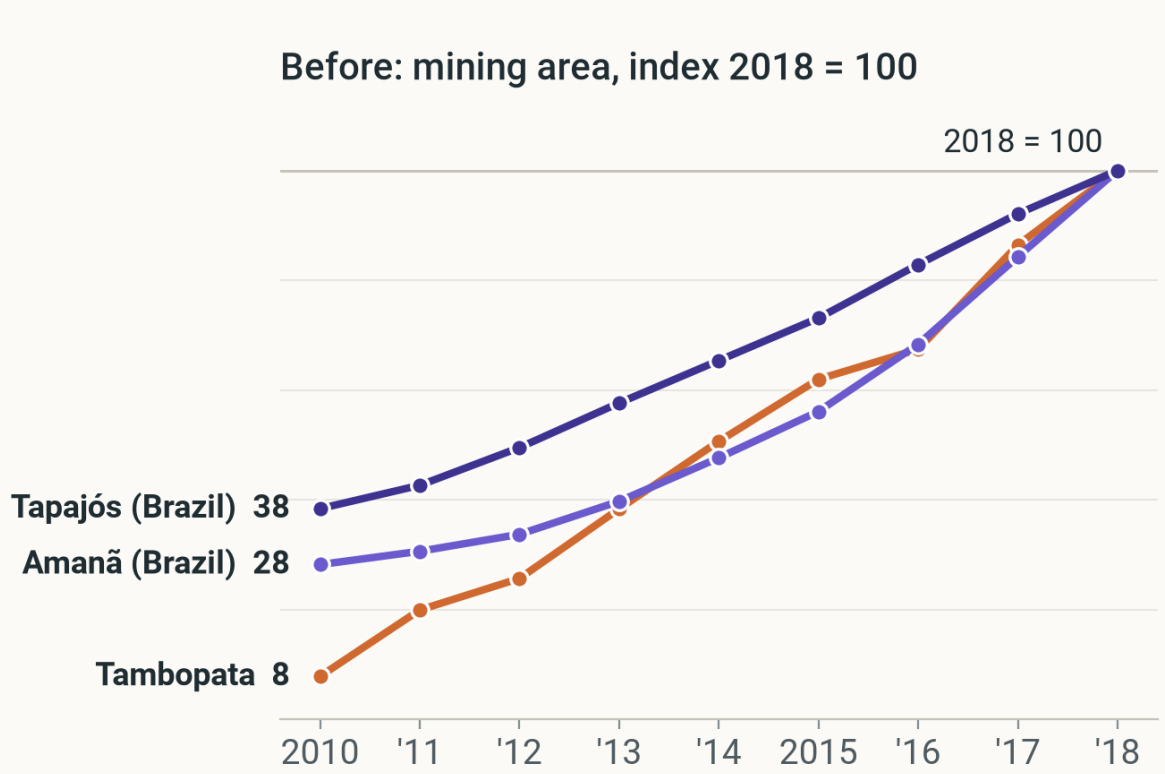


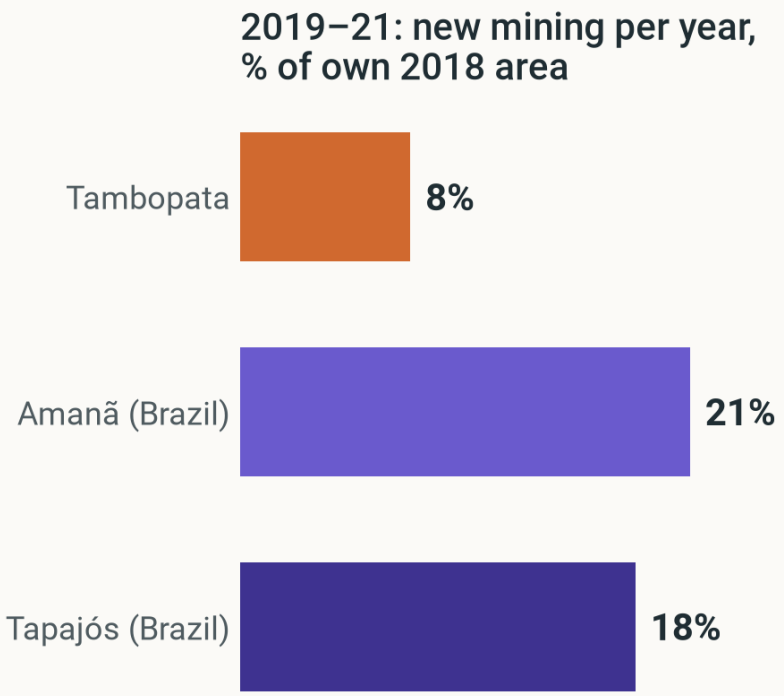
Source: MapBiomas Peru Col 4 rasters, zones v2. Logit on 1-km cells: distance to highway and rivers, gold-bearing land, mining nearby, zone. Trained on 2018 → 2019–21, tested on 2022–25 in held-out 10-km blocks, applied to 2025 features. It misses the 2025 reserve incursion (river access, not in the model). Prediction, not a causal map.

DESCRIPTION

The Brazilian twins fail the pre-trend test

Mining area before 2019 (index 2018 = 100) · new mining 2019–21, % of 2018 area





Source: MapBiomas Peru Col 4 (Tambopata) and MapBiomas Amazonía Col 6 (twins) before 2019; Amazon Mining Watch (480 m) for 2019–21. Different collections: compared as indices, never subtracted. Twins chosen by fixed rules (control_select.py).

DESCRIPTION

Brazil: falls, but not after each operation

YANOMAMI, FIRST-TIME MINING

518 → 458

ha/yr, 2019–22 → 2023–25

STAGGERED DID, 24 TERRITORIES

+0.26

log points, p = 0.44: wrong sign

NATIONAL REFORMS, JAN 2023

All at once

new rules on gold sales hit every territory

Consistent with 'enforcement plus gold tracing goes with less mining'. Our test can't confirm it.

DESCRIPTION

The five map layers as one video

